



2. 5. How to write Linux image to SPI Flash+NVMe SSD

Note that the Linux image mentioned here specifically refers to the images of Linux distributions such as Debian or Ubuntu downloaded from the Orange Pi data download page.

Note that all the following operations are performed on a Windows computer.

2. 5. 1. How to use RKDevTool to burn

1) First, you need to prepare an NVMe SSD. The PCIe supported by the M.2 slot of the development board is PCIe2.0x1, and the theoretical maximum speed is 500MB/s. PCIe3.0 and PCIe4.0 NVMe SSDs are also available, but the highest speed is only PCIe2.0.

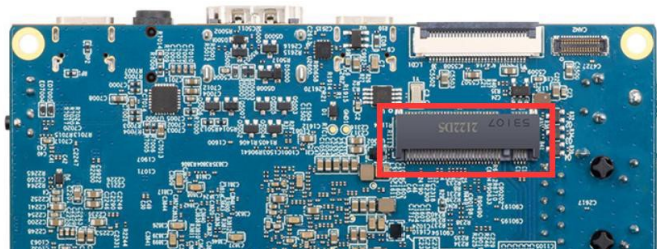
a. The M.2 2230 SSD is as follows



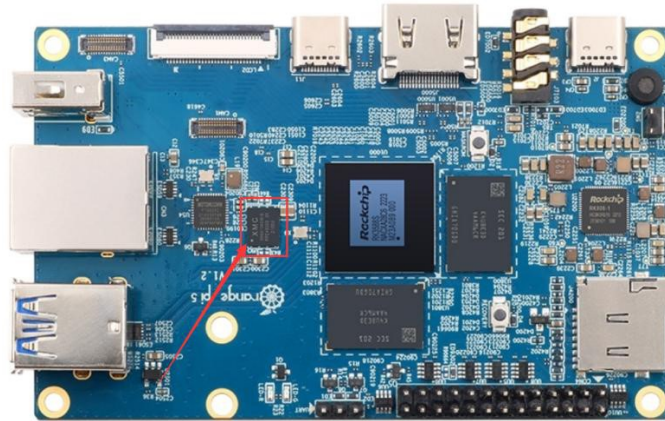
b. The M.2 2242 SSD is as follows



2) Then insert the NVMe SSD into the M.2 PCIe interface of the development board and fix it



3) The position of the SPI Flash on the development board is shown in the figure below, no other settings are required before starting the programming

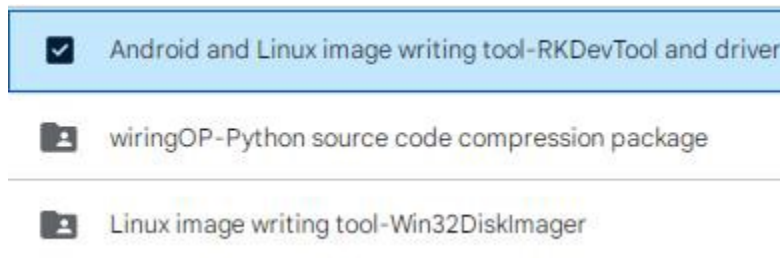


4) Then you need to prepare a data cable with good quality Type-C interface

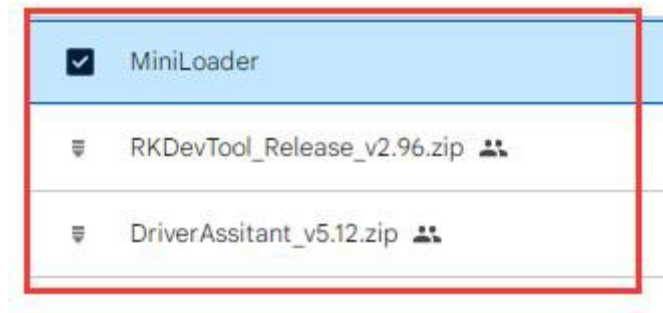


5) Then download Rockchip **DriverAssitant_v5.12.zip** and **MiniLoader** and the burning tool **RKDevTool_Release_v2.96.zip** from the [Orange Pi data download page](#), please make sure that the version of the downloaded **RKDevTool** tool is **v2.96**

- a. On the Orange Pi data download page, first select the official tool, and then enter the following folder



- b. Then download all the files below



Note that the folder of MiniLoader - the things needed to burn the Linux image is hereinafter referred to as the MiniLoader folder.

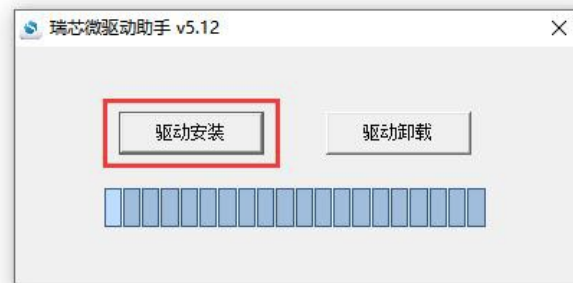
5) Then download the Linux operating system image file compression package that you want to burn from the [Orange Pi data download page](#), and then use the decompression software to decompress it. Among the decompressed files, the file ending with ".img" is the image file of the operating system , the size is generally above 2GB

6) Then use decompression software to decompress **DriverAssitant_v5.12.zip**, and then find the **DriverInstall.exe** executable file in the decompressed folder and open it

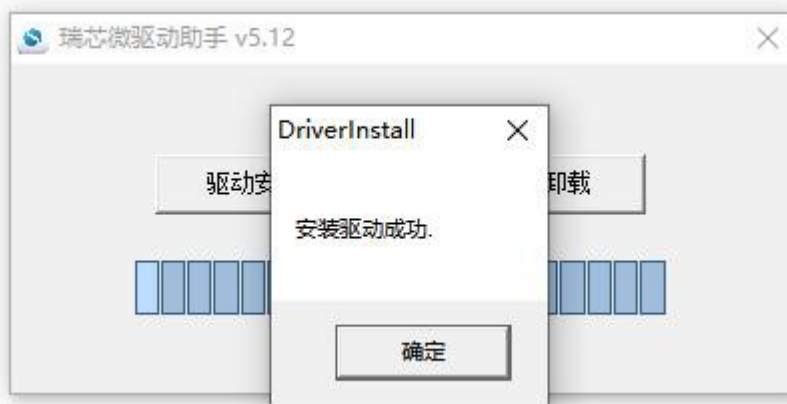
名称	修改日期	类型	大小
ADBDriver	2022/12/1 15:07	文件夹	
bin	2022/12/1 15:07	文件夹	
Driver	2022/12/1 15:07	文件夹	
config	2014/6/3 15:38	配置设置	1 KB
DriverInstall	2022/2/28 14:11	应用程序	491 KB
Readme	2018/1/31 17:44	文本文档	1 KB
revision	2022/2/28 14:14	文本文档	1 KB

7) After opening **DriverInstall.exe**, the steps to install the Rockchip driver are as follows

- a. Click the "**Driver Installation**" button



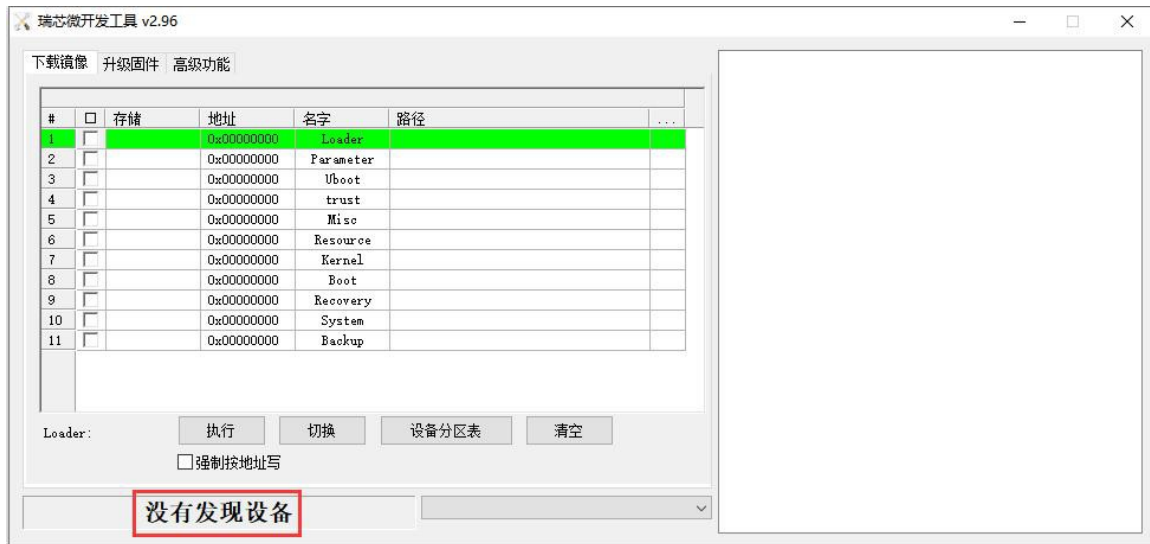
- b. After waiting for a while, a pop-up window will prompt "**driver installed successfully**", and then click the "**OK**" button.



- 8) Then decompress **RKDevTool_Release_v2.96.zip**, this software does not need to be installed, just find **RKDevTool** in the decompressed folder and open it

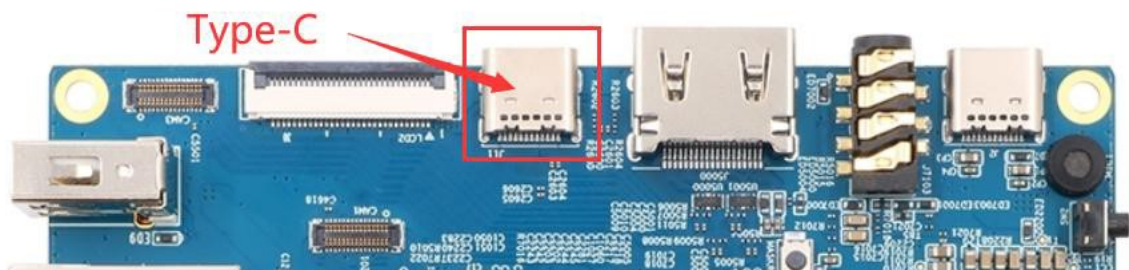
名称	修改日期	类型	大小
bin	2022/12/1 15:07	文件夹	
Language	2022/12/1 15:07	文件夹	
config.cfg	2022/3/23 9:11	CFG 文件	7 KB
config	2021/11/30 11:04	配置设置	2 KB
revision	2022/5/27 9:09	文本文档	3 KB
RKDevTool	2022/5/27 9:06	应用程序	1,212 KB
开发工具使用文档_v1.0	2021/8/27 10:28	Foxit PDF Reade...	450 KB

- 9) After opening the **RKDevTool** burning tool, because the computer has not been connected to the development board through the Type-C cable at this time, the lower left corner will prompt "**No device found**"

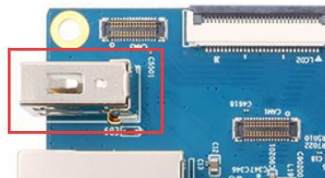


10) Then start burning the Linux image to the SSD

- a. First, connect the development board to the Windows computer through the Type-C data cable. The position of the Type-C interface on the development board is shown in the figure below.



- b. Make sure that the development board is not connected to the power supply and inserted into the TF card.
- c. Also make sure that the white USB2.0 interface in the position shown below is not plugged into a USB device USB2.0.



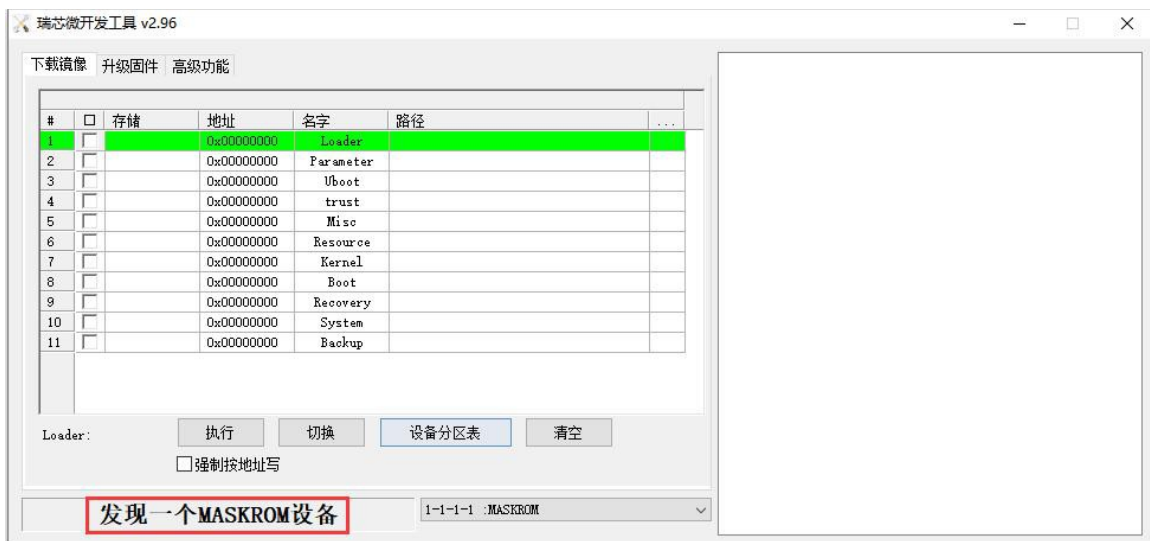
- d. Then press and hold the MaskROM button on the development board. The position of the MaskROM button on the development board is shown in the figure below:



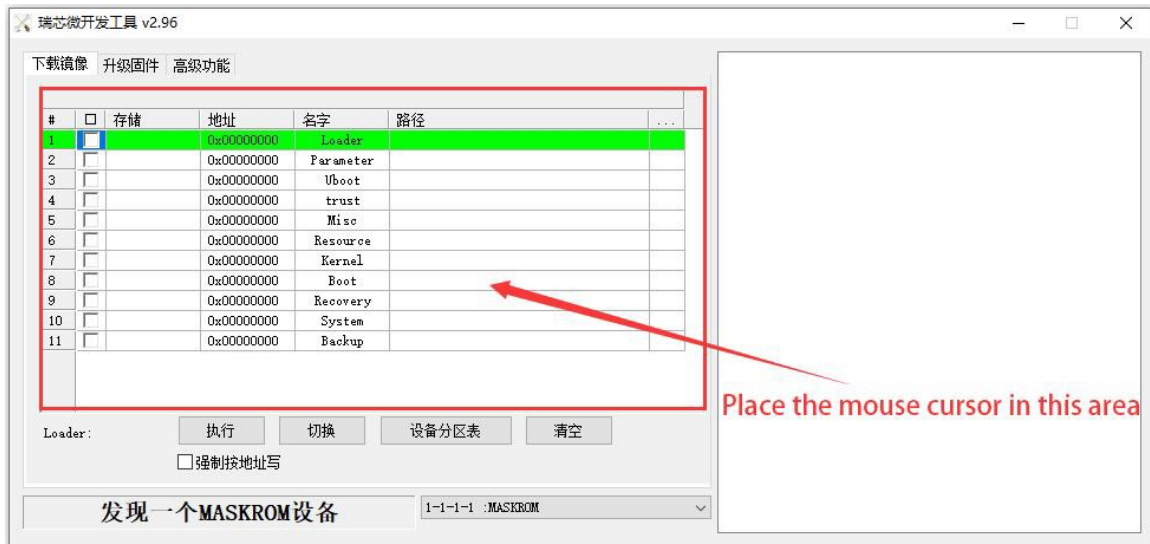
- e. Then connect the power supply of the Type-C interface to the development board, and power on, and then release the MaskROM button



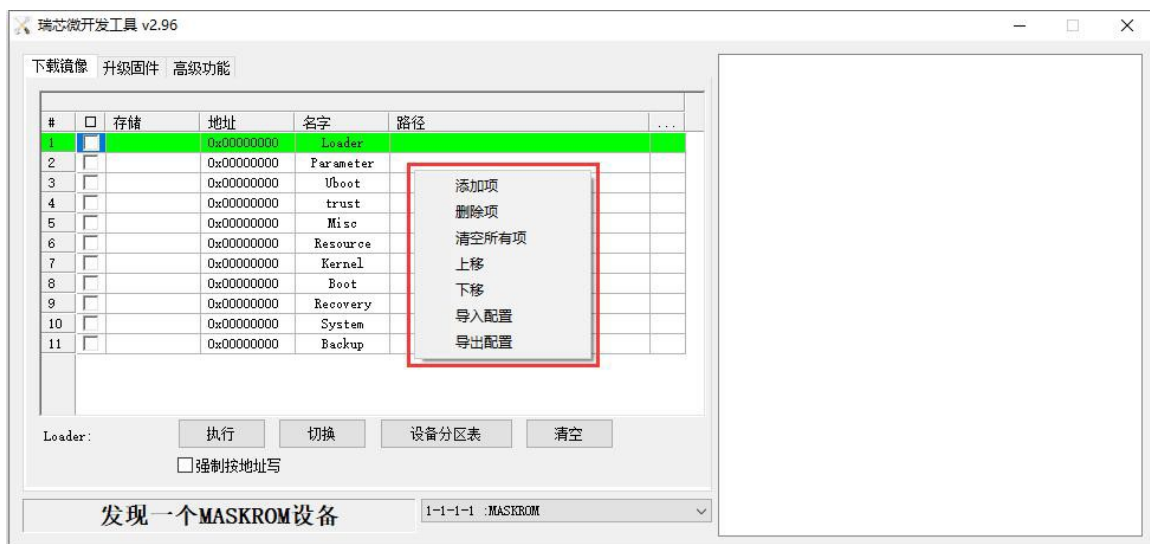
- f. If the previous steps are successful, the development board will enter the **MASKROM** mode at this time, and the interface of the burning tool will prompt "found a MASKROM device"



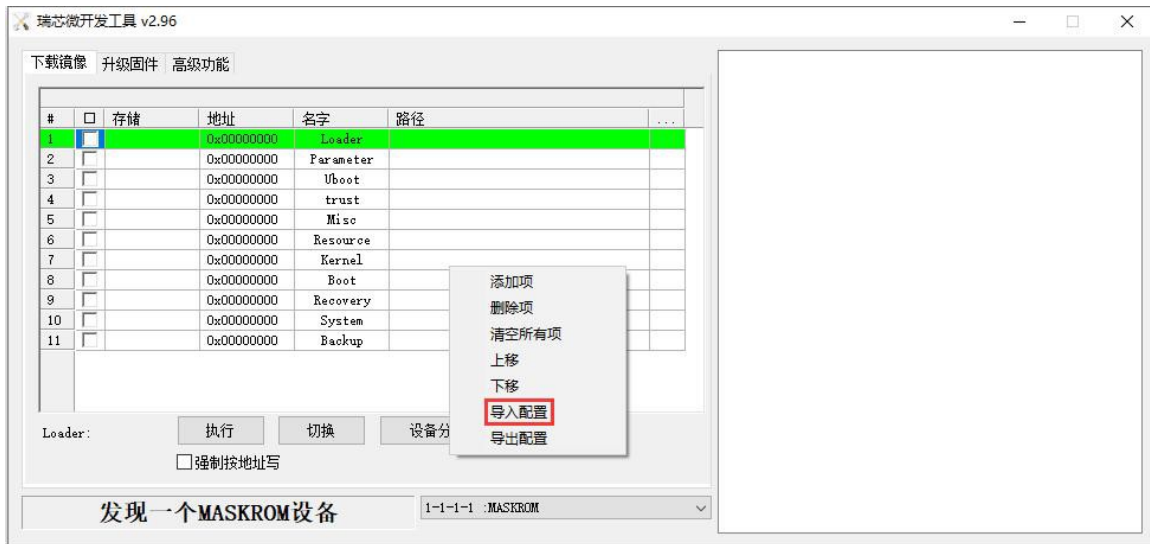
- g. Then place the mouse cursor in the area below



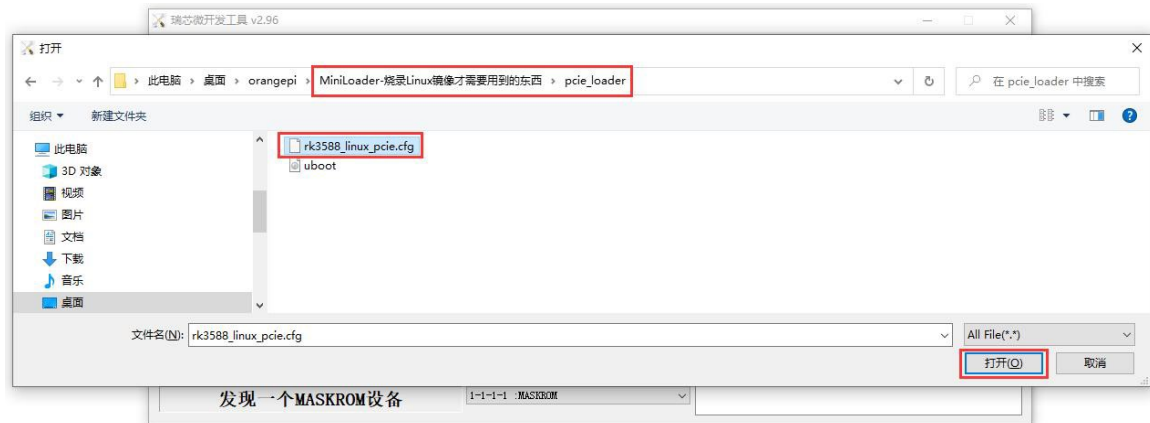
- h. Then click the right mouse button and the selection interface shown in the figure below will pop up



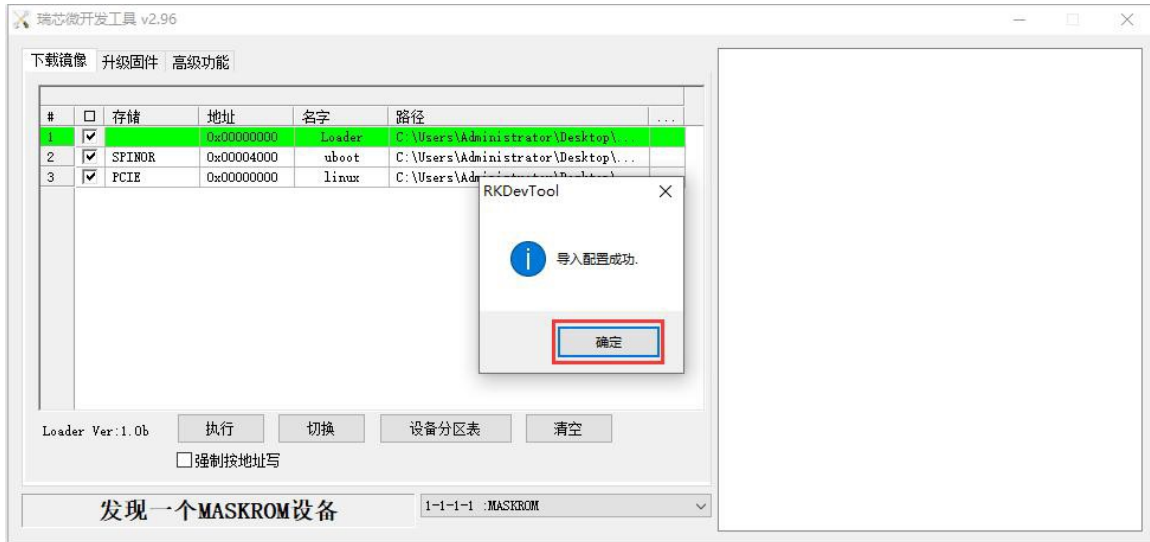
- i. Then select the **import configuration** option



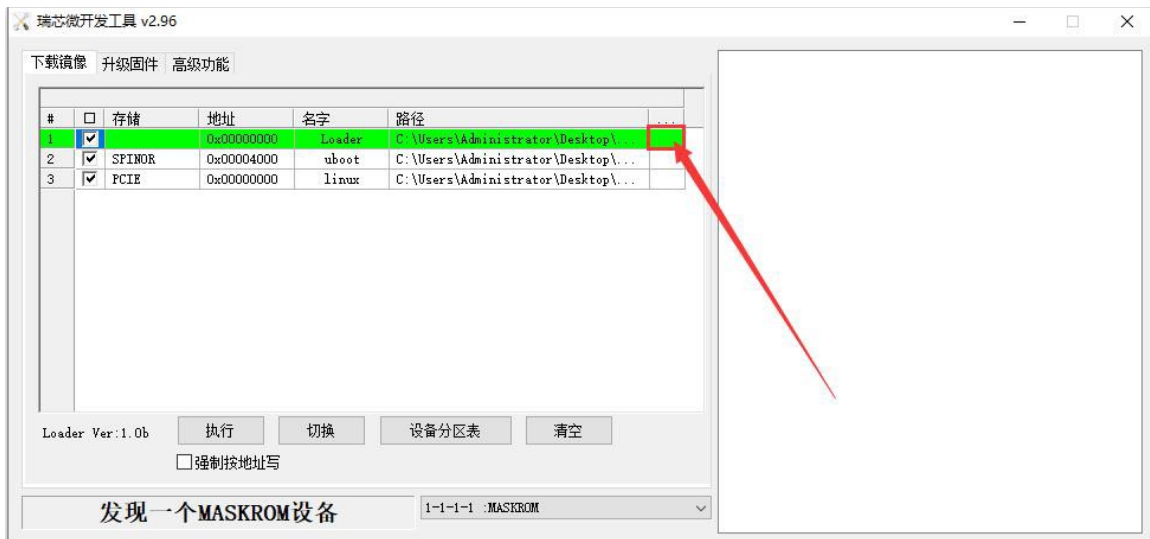
- j. Then enter the **MiniLoader** folder downloaded earlier, and then enter the pcie_loader folder, then select the **rk3588_linux_pcie.cfg** configuration file, and click **Open**.



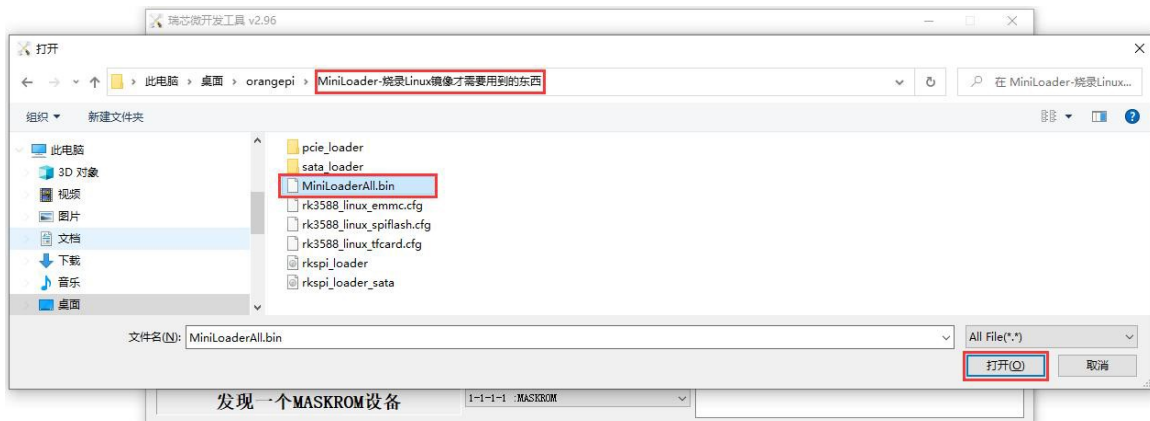
- k. Then click **OK**



1. Then click the position shown in the figure below

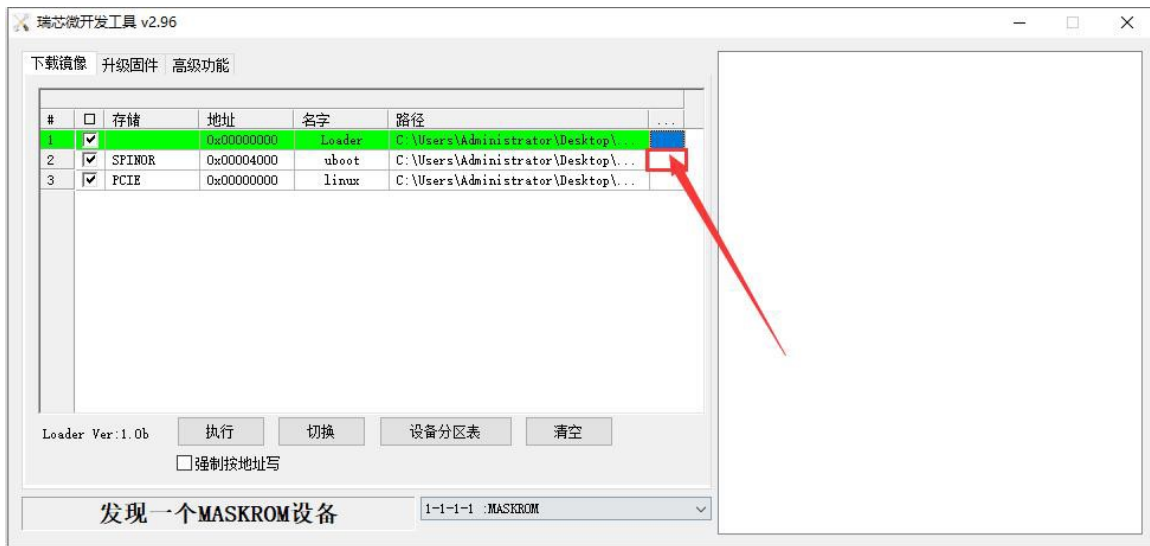


m. Select **MiniLoaderAll.bin** in the **MiniLoader** folder downloaded earlier, and then click to open

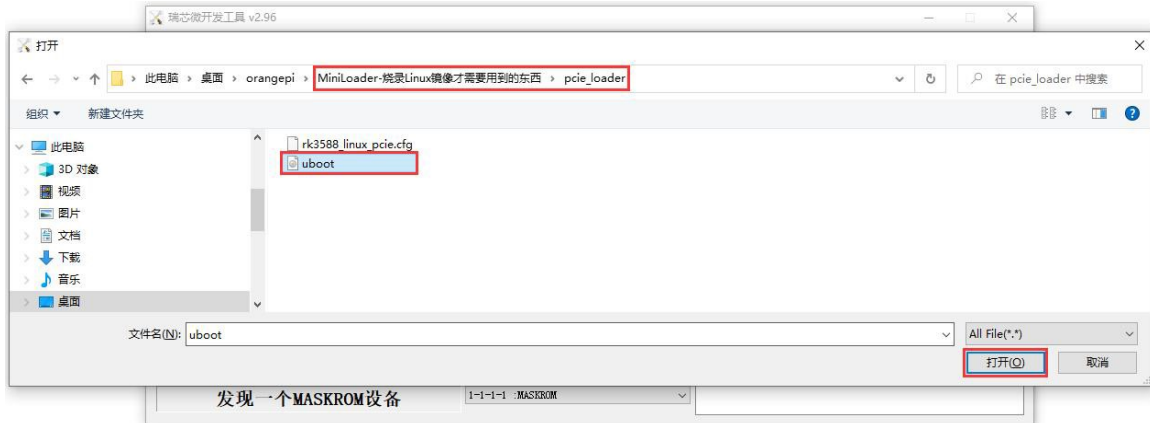




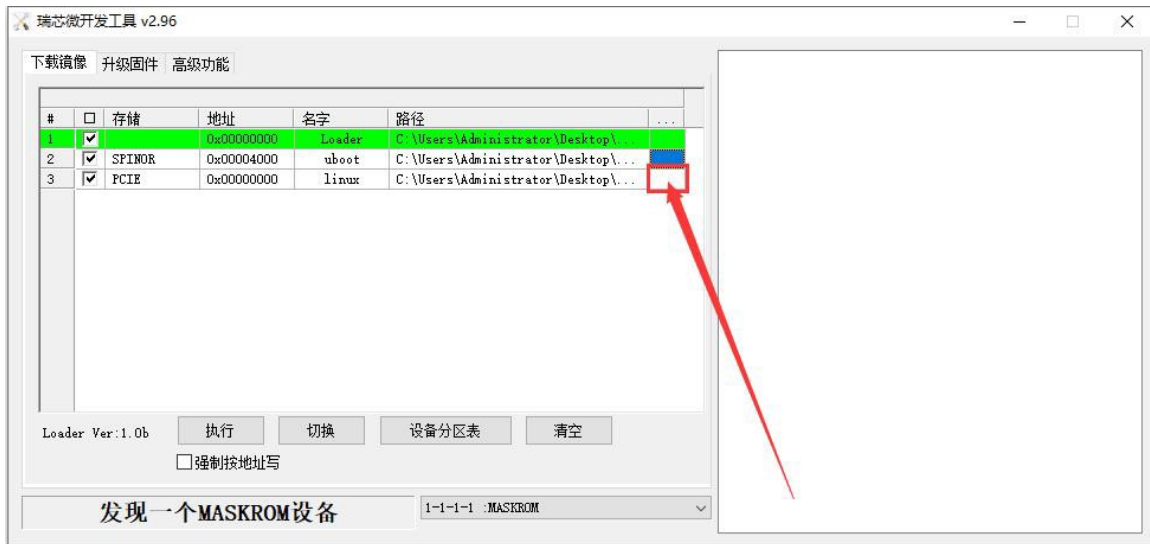
n. Then click the position shown in the figure below



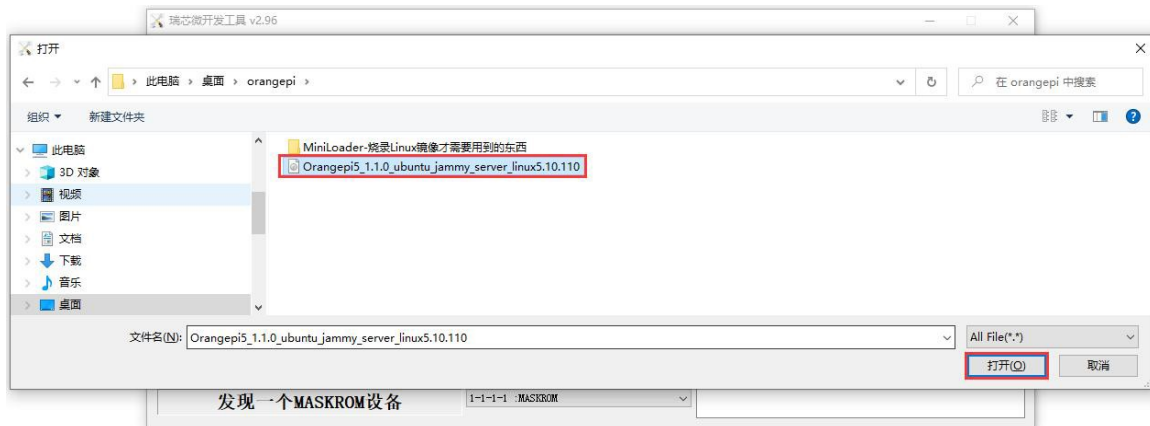
o. Then enter the **MiniLoader** folder downloaded earlier, then enter the **pcie_loader** folder, then select the **uboot** file, and click **Open**.



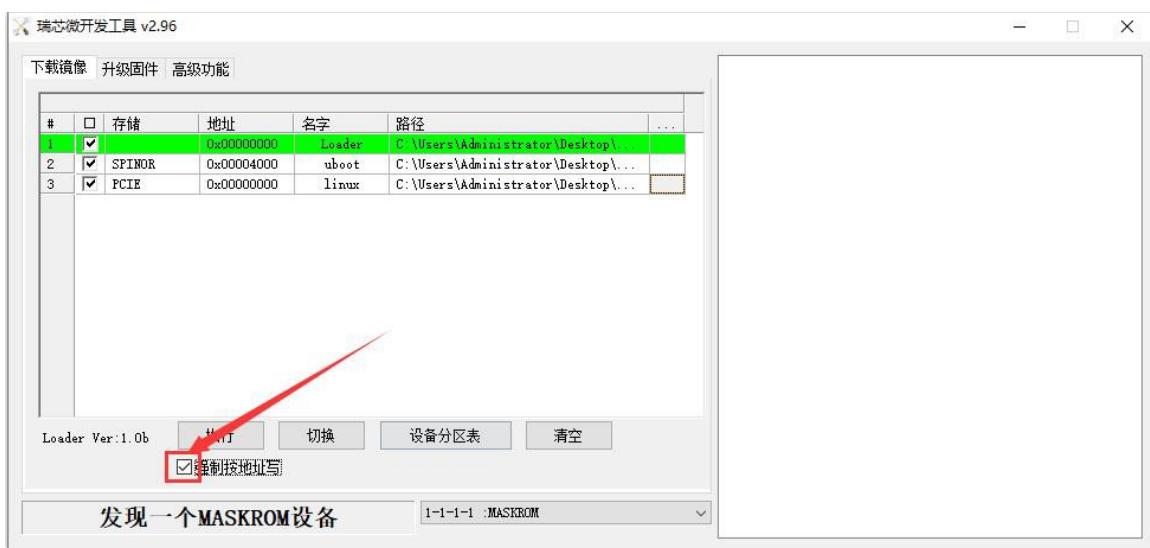
p. Then click the position shown in the figure below



q. Then select the path of the linux image you want to burn, and click **Open**

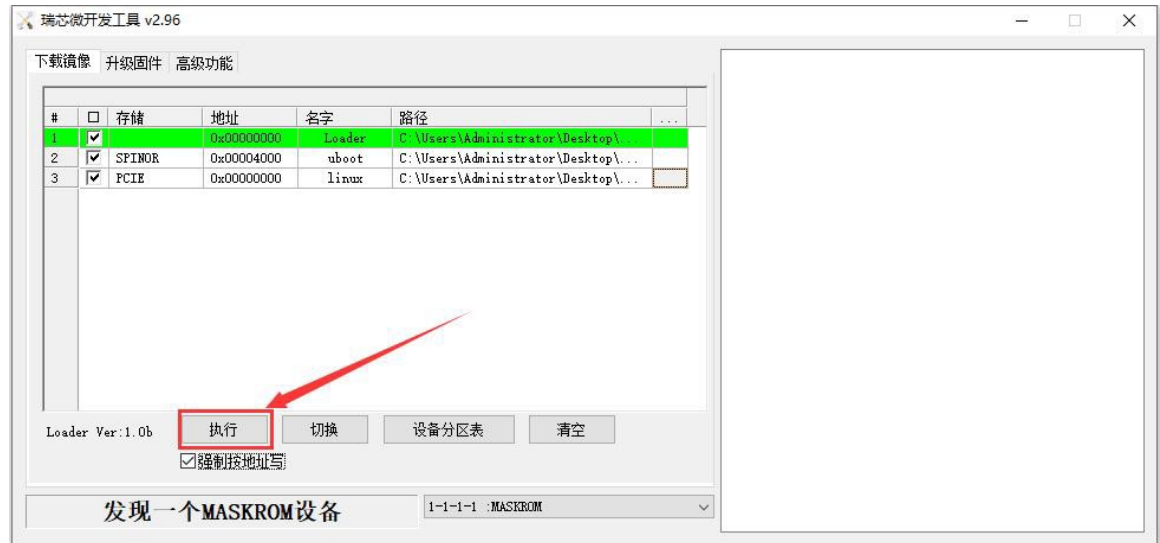


r. Then please check the option to **force writing by address**

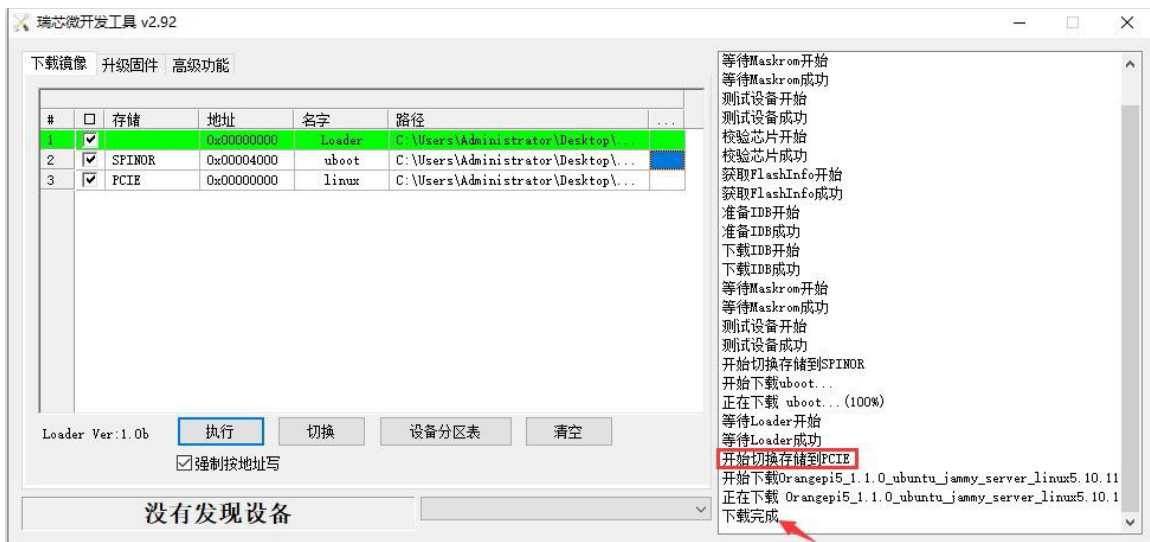




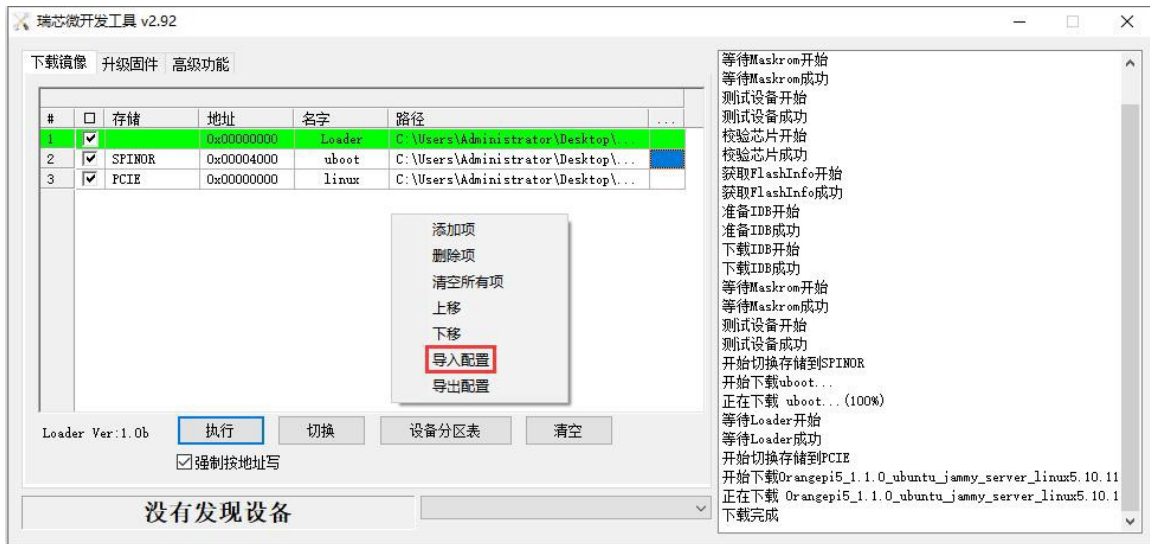
- s. Click the Execute button again to start burning the linux image to the SSD



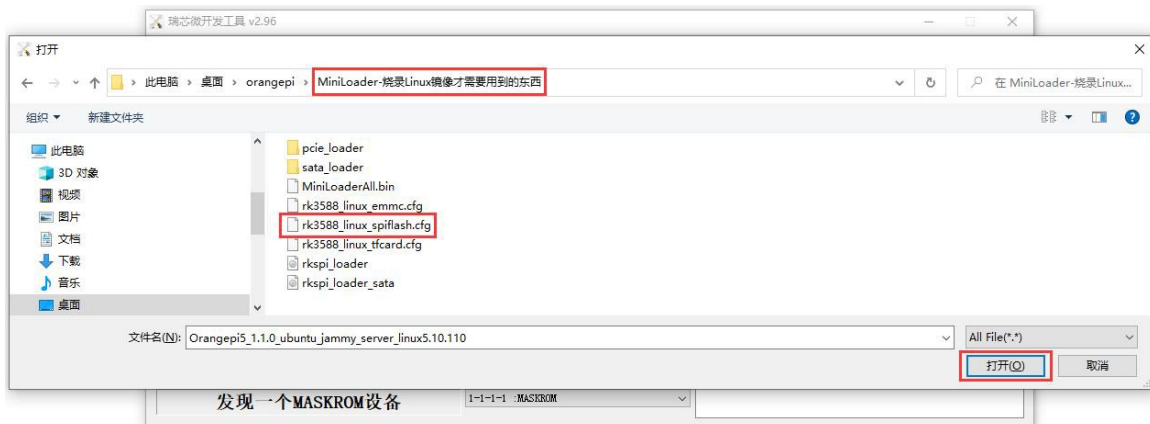
- t. The log displayed after burning the linux image is shown in the figure below



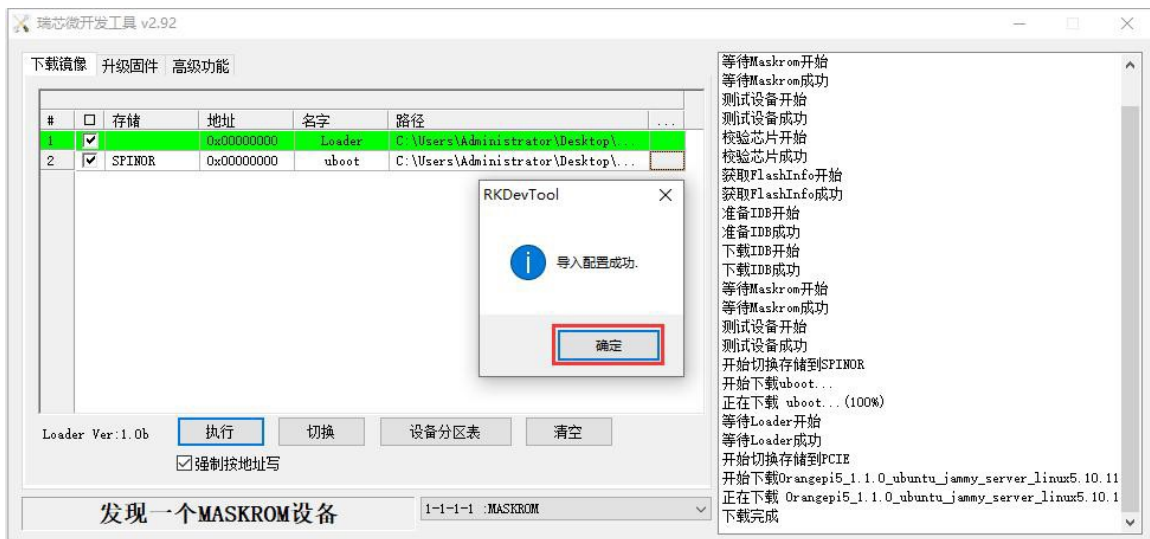
- u. Then burn the u-boot image required by the linux system in the SSD to SPIFlash, and **first enter the MaskROM mode again.**
- v. Then select the **import configuration** option



- w. Then select the **rk3588_linux_spiflash.cfg** configuration file in the **MiniLoader** folder downloaded earlier, and click **Open**.

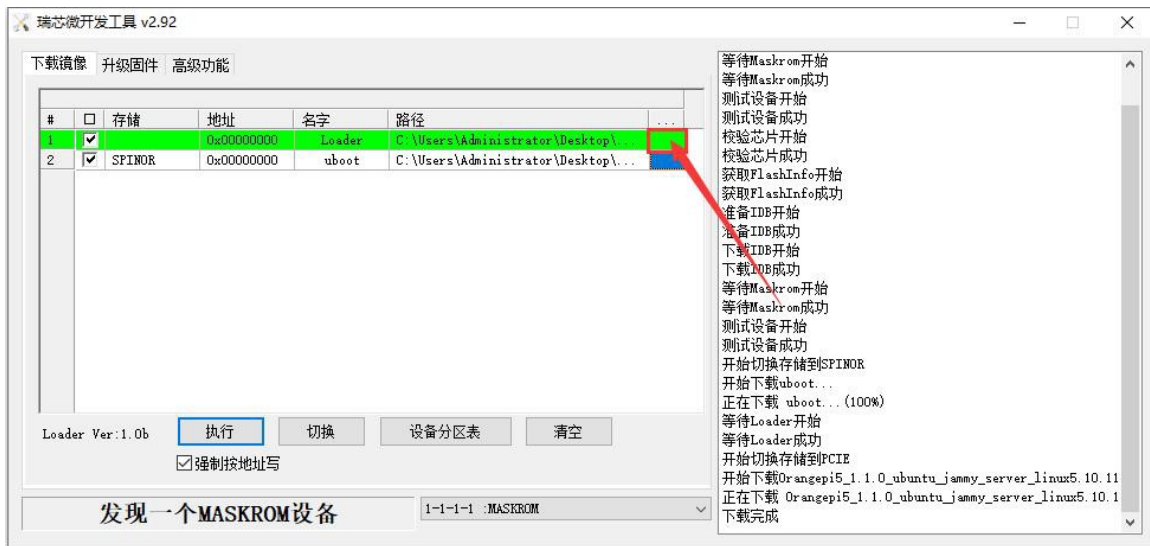


- x. Then click **OK**

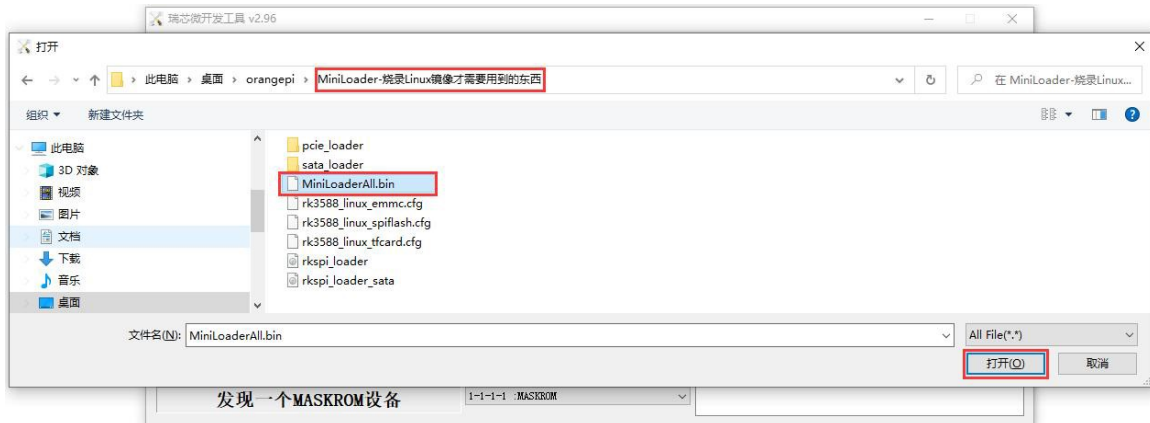




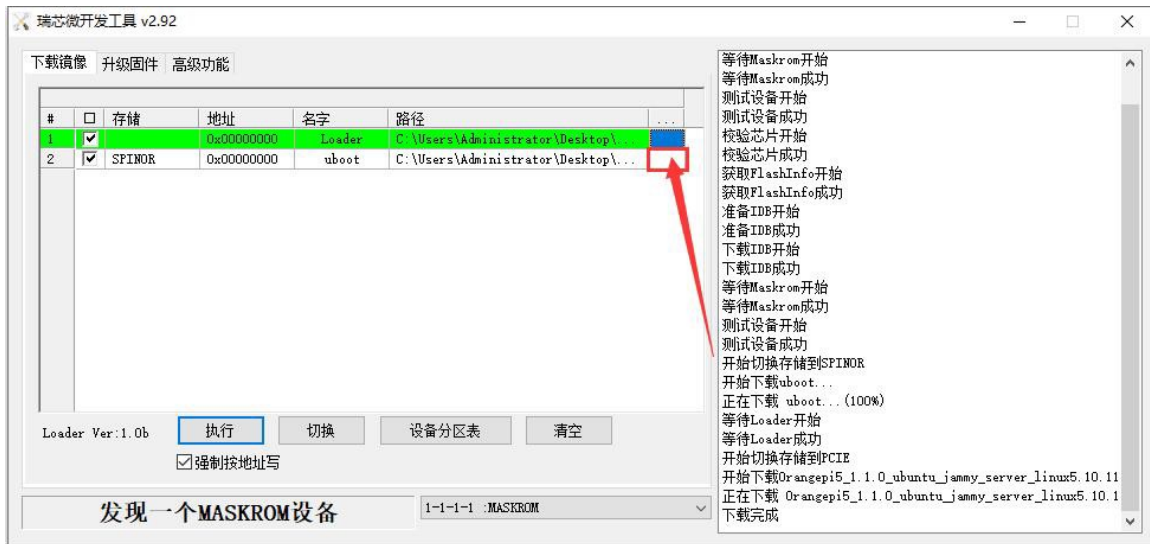
y. Then click the position shown in the figure below



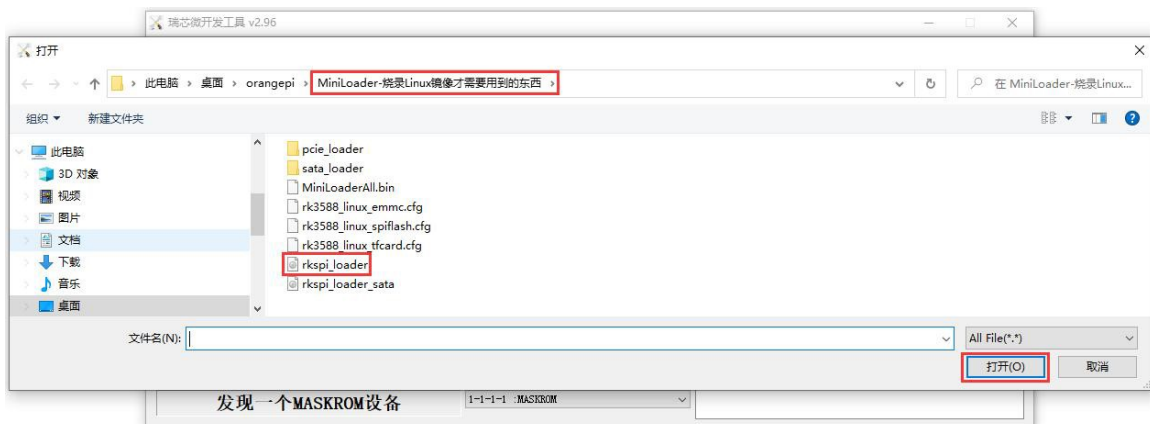
z. Select **MiniLoaderAll.bin** in the **MiniLoader** folder downloaded earlier, and then click to open



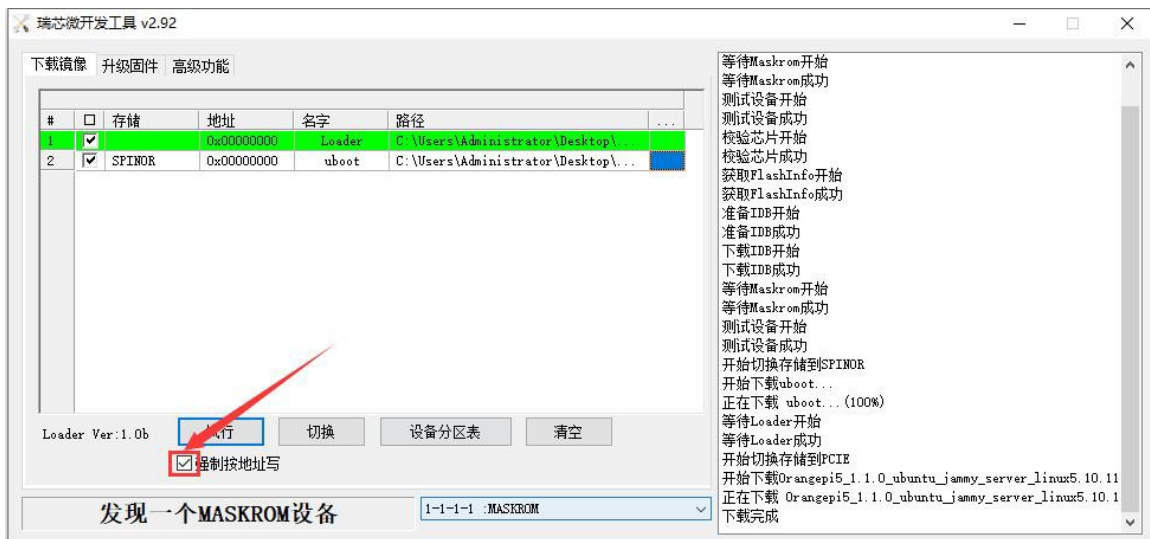
aa. Then click the location shown in the figure below



ab. Then select **rkspi_loader** in the **MiniLoader** folder downloaded earlier, and then click to **open**

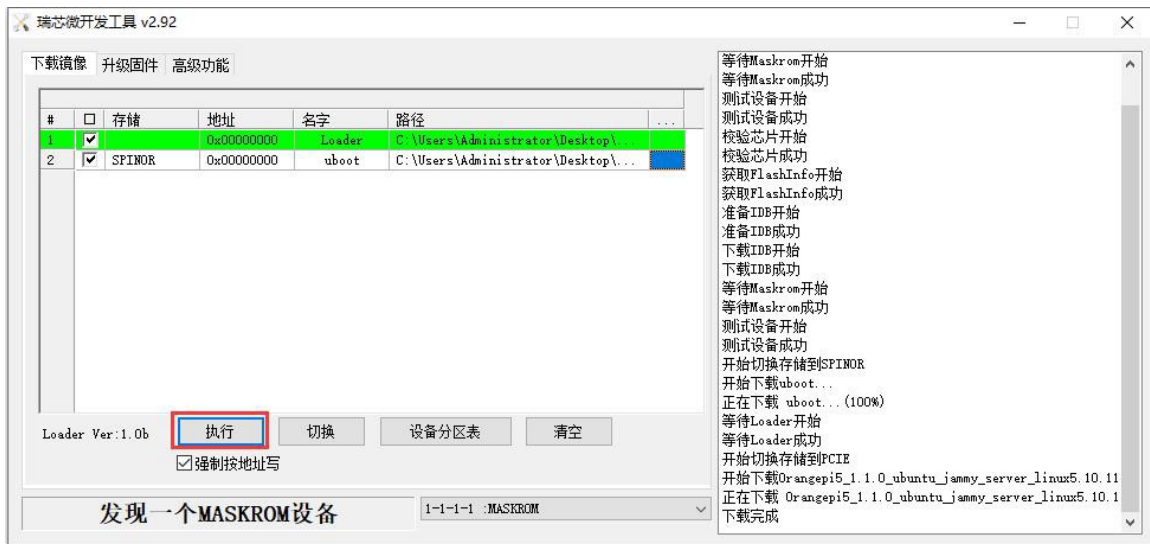


ac. Then please make sure that the option to **force writing by address** is ticked

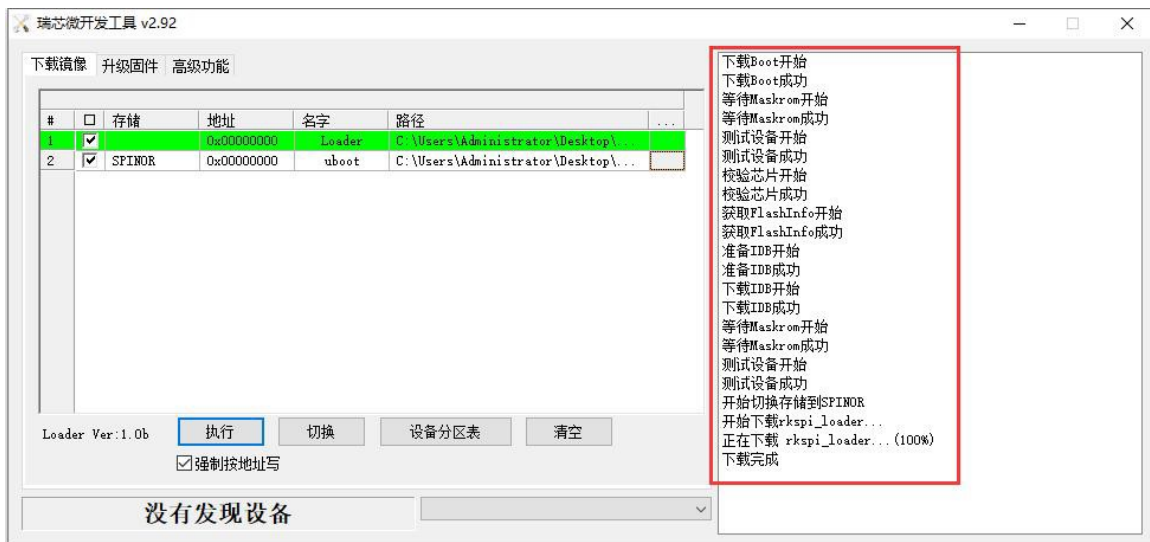




ad. Click the Execute button again to start burning the u-boot image into SPIFlash



ae. The display log after burning the u-boot image is shown in the figure below.



af. After the u-boot image is burnt, it will automatically start the linux system in the SPIFlash+PCIe SSD. If it does not start normally, please power on and try again.

2. 5. 2. How to use the dd command to burn

1) First, you need to prepare an NVMe SSD. The PCIe supported by the M.2 slot of the development board is PCIe2.0x1, and the theoretical maximum speed is 500MB/s. PCIe3.0 and PCIe4.0 NVMe SSDs are also available, but the highest speed is only PCIe2.0x1 PCIe2.0x1.

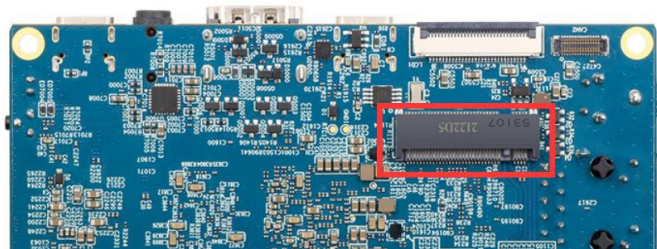
a. The M.2 2230 SSD is as follow



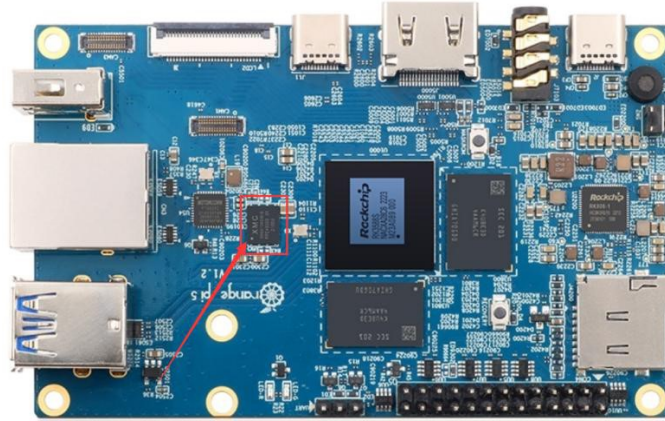
b. The M.2 2242 SSD is as follow



2) Then insert the NVMe SSD into the M.2 PCIe interface of the development board and fix it



3) The position of the SPI Flash on the development board is shown in the figure below, no other settings are required before starting



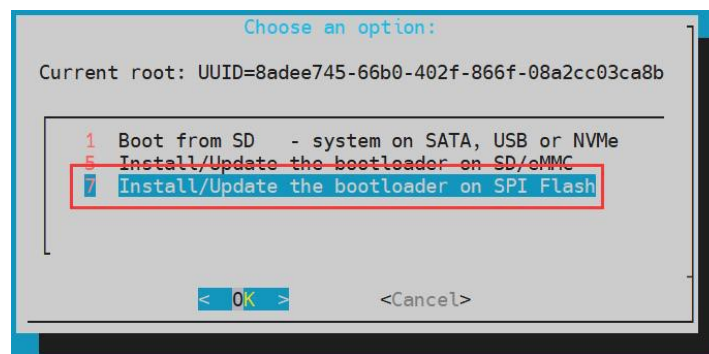
4) Burning the linux image to SPIFlash+NVMe SDD needs to be completed with the help of a TF card, so first you need to burn the linux image to the TF card, and then use the TF card to start the development board to enter the linux system. For the method of burning the Linux image to the TF card, please refer to the instructions in the two sections of [the method of burning the Linux image to the TF card based on the Windows PC](#) and [the method of burning the Linux image to the TF card based on the Ubuntu PC](#)

5) After using the TF card to start the Linux system, we first burn the u-boot image into the SPI Flash

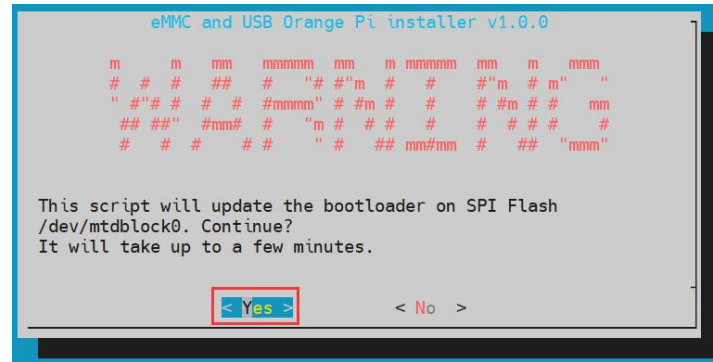
a. Run **nand-sata-install** first, **ordinary users remember to add sudo permission**

```
orange@orange:~$ sudo nand-sata-install
```

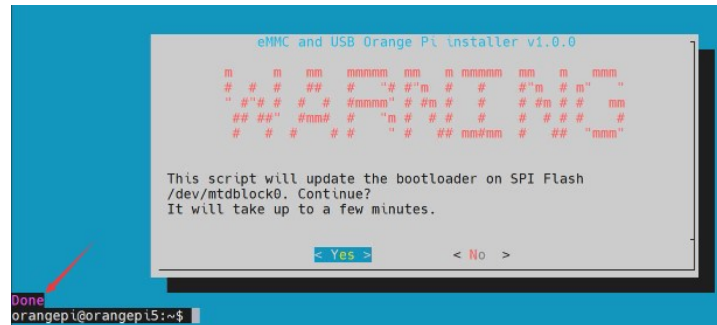
b. Then select **7 Install/Update ther bootloader on SPI Flash**



c. Then select **<Yes>**



- d. Then please wait patiently for the burning to complete. After the burning is completed, the display will be as follows (a **Done** will be displayed in the lower left corner)



There is no nand-sata-install script in OPi OS Arch system, please use the following command to image u-boot to SPI Flash:

```
[orangepi@orangepi ~]$ sudo dd if=/boot/rkspi_loader.img of=/dev/mtdblock0
```

6) Then upload the linux image file (Debian or Ubuntu image downloaded from the official website) to the TF card. For the method of uploading the linux image file to the development board, please refer to the description in the section of [the method of uploading files to the development board Linux system](#)

7) After uploading the image to the linux system of the development board, we enter the storage path of the image file in the command line of the linux system of the development board. For example, I store the linux image of the development board in the **/home/orangepi/Desktop** directory. Download it, and then enter the **/home/orangepi/Desktop** directory to see the uploaded image file

```
orangepi@orangepi:~$ cd /home/orangepi/Desktop
orangepi@orangepi:~/Desktop$ ls
```



```
OrangePi5_x.x.x_debian_bullseye_desktop_xfce_linux5.10.110.img
```

How to enter the command line of the development board linux system?

1. For the method of using the serial port to log in to the terminal, please refer to the instructions in the section on [how to use the debugging serial port](#)
2. Use ssh to remotely log in to the Linux system, please refer to the instructions in the section of [SSH remote login to the development board](#).
3. If HDMI, LCD and other display screens are connected, you can open a command line terminal on the desktop.

8) Next, let's confirm that the NVMe SSD has been recognized by the development board's linux. If the NVMe SSD is recognized normally, use the **sudo fdisk -l** command to see **nvme** related information.

```
orangePi@orangePi:~/Desktop$ sudo fdisk -l | grep "nvme0n1"
Disk /dev/nvme0n1: 1.86 TiB, 2048408248320 bytes, 4000797360 sectors
```

Use the **lspci** command to see an NVMe-related PCI device

```
orangePi@orangePi:~/Desktop$ lspci
0004:40:00.0 PCI bridge: Fuzhou Rockchip Electronics Co., Ltd Device 3588 (rev 01)
0004:41:00.0 Non-Volatile memory controller: MAXIO Technology (Hangzhou) Ltd.
NVMe SSD Controller MAP1202 (rev 01)
```

9) Then we can use the dd command to clear the NVMe SSD(Optional)

```
orangePi@orangePi5:~/Desktop$ sudo dd bs=1M if=/dev/zero of=/dev/nvme0n1 count=2000 status=progress
orangePi@orangePi5:~/Desktop$ sudo sync
```

10) Then you can use the dd command to burn the linux image of the development board to the NVMe SSD.

- a. 下面 In the following command, the **if=** parameter is followed by the full path where the linux image is stored + the name of the Linux image (such as **the name of /home/orangepi/Desktop/Linux image**). Because we have entered the path of the linux image above, we only need to fill in the name of the Linux image.
- b. Please do not copy the linux image name in the following command, but replace it with the actual image name (because the version number of the image may be



updated)

```
sudo dd bs=1M if=OrangePi5_x.x.x_debian_bullseye_desktop_xfce_linux5.10.110.img of=/dev/nvme0n1 status=progress
```

```
sudo sync
```

Note, if you upload a .7z or .xz linux image compressed file, please remember to decompress it before using the dd command to burn.

The detailed description of all parameters of the dd command and more usage can be viewed by executing the man dd command in the linux system.

11) After successfully burning the linux image of the development board to the NVMe SSD, you can use the poweroff command to shut down. Then please pull out the TF card, and then short press the power button to turn on, and then the linux system in SPIFlash+NVMe SSD will be started.

12) After starting the system in the NVMe SSD, use the **df -h** command to see the actual hard disk capacity NVMe SSD.

a. 128GB NVMe SSD

```
orangepi@orangepi:~$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	3.8G	8.0K	3.8G	1%	/dev
tmpfs	769M	1.4M	768M	1%	/run
/dev/nvme0n1p2	118G	5.8G	111G	5%	/
tmpfs	3.8G	0	3.8G	0%	/dev/shm
tmpfs	5.0M	4.0K	5.0M	1%	/run/lock
tmpfs	3.8G	16K	3.8G	1%	/tmp
/dev/nvme0n1p1	256M	90M	166M	36%	/boot
/dev/zram1	194M	9.9M	170M	6%	/var/log
tmpfs	769M	60K	769M	1%	/run/user/1000
tmpfs	769M	48K	769M	1%	/run/user/0

b. 2TB NVMe SSD

```
orangepi@orangepi:~$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	3.8G	8.0K	3.8G	1%	/dev



tmpfs	769M	1.4M	768M	1% /run
/dev/nvme0n1p2	1.9T	4.1G	1.8T	1% /
tmpfs	3.8G	0	3.8G	0% /dev/shm
tmpfs	5.0M	4.0K	5.0M	1% /run/lock
/dev/zram2	3.7G	76K	3.5G	1% /tmp
/dev/nvme0n1p1	256M	90M	166M	36% /boot
/dev/zram1	194M	15M	165M	9% /var/log
tmpfs	769M	60K	769M	1% /run/user/1000
tmpfs	769M	48K	769M	1% /run/user/0

13) When the TF card and NVMe SSD are programmed with exactly the same system, **if both the TF card and NVMe SSD are inserted into the development board, power on the development board at this time, and u-boot will give priority to starting the system in the TF card.** However, since the systems in the TF card and NVMe SSD are exactly the same, the UUIDs of the **/boot** partition and the **rootfs** partition in the two storage devices are also the same, which may cause the partition in the NVMe SSD to be loaded when the TF card starts. Running the script below resolves this issue

```
orange@orange:~$ sudo fix_mmc_ssd.sh
```

Exactly the same system means that the image name is exactly the same. Even if they are all Debian11 systems, the versions are different

There is no fix_mmc_ssd.sh script in OPi OS Arch system.

2.5.3. How to use the balenaEtcher to burn

Please do not use this method for OPi OS Arch system.

1) First, you need to prepare an NVMe SSD. The PCIe supported by the M.2 slot of the development board is PCIe2.0x1, and the theoretical maximum speed is 500MB/s. PCIe3.0 and PCIe4.0 NVMe SSDs are also available, but the highest speed is only PCIe2.0x1.

a. The M.2 2230 SSD is as follows